OCR
Oxford Cambridge and RSA

Tuesday 7 June 2022 – Afternoon

A Level Mathematics B (MEI)

H640/01 Pure Mathematics and Mechanics

Printed Answer Booklet

Time allowed: 2 hours

912045115 11

You must have:
- Question Paper H640/01 (inside this document)
- a scientific or graphical calculator

H640 01

Please write clearly in black ink. **Do not write in the barcodes.**

Centre number | | | | | | Candidate number | | | | |

First name(s) _____

Last name _____

INSTRUCTIONS
- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided in the **Printed Answer Booklet**. If you need extra space use the lined pages at the end of the Printed Answer Booklet. The question numbers must be clearly shown.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.
- Give your final answers to a degree of accuracy that is appropriate to the context.
- The acceleration due to gravity is denoted by $g\,\text{m}\,\text{s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

INFORMATION
- This document has **20** pages.

ADVICE
- Read each question carefully before you start your answer.

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Section A (24 marks)

1(a)	
1(b)	
2	

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3(a)

3(b)

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4	

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5(a)

5(b)

Tension in the string =

Value of F =

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Section B (76 marks)

6(a)

6(b)

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6(c)

6(d)

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7(a)

7(b)

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8(a)

8(b)

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8(c)

8(d)

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9(a)

9(b)

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9(c)

10

(answer space continued on next page)

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10 **(continued)**

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11

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12

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13(a)

13(b)

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13(c)

13(d)

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14(a)

14(b)

14(c)

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14(d)	
14(e)	
	Time =
	Temperature =

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ADDITIONAL ANSWER SPACE

If additional space is required, you should use the following lined page(s). The question number(s) must be clearly shown in the margin(s).

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